Cumulative learning through acquired observation operations

A self-contained Astra investigation · 31 October 2026

Result. A learner can acquire exact executable functions from noisy observations, then use those functions to align a different collection of noisy tasks into repeated observations of a new unknown function. The relationships, useful numerical noise bounds, and task programs are acquired. The final construction receives no hidden graph, task grouping, planted rank, or numerical noise rate.

The mechanism is conditional on a demanding but explicit observation model: uniform binary inputs, parity tasks, and independently corrupted channel labels at the same input. It includes a procedure that can decline when the observed evidence is insufficient. Its current-episode correctness does not require past task associations to have been correct: fresh tests assess the actual executable functions.

Executed stage	Exact task programs	Acquired rank	Noisy labels
Initial acquisition	276 / 276	23	439,392
Later acquisition	21 / 21	+1	33,453
Complete pipeline	297 / 297	24 total	472,845

The complete pipeline has a declared source-conditional failure bound of 0.03. One independent deterministic replay reproduced both frozen models and every observation array exactly. These are executable and mathematical checks, not an empirical estimate of failure probability or evidence that a real source meets the assumptions.

Scientific status

The contribution is a fully specified composition and organizing principle, with ordinary proofs, exact finite decisions, code, and counterexamples. Its components have close established predecessors. Historical priority of the complete assembly is unresolved. An ordinary retained-data learner also certifies every final task at the same confidence objective; a practical full-cost advantage has not been demonstrated.

This investigation starts from accessible sources. An earlier predictive compiler and its completed checks are preserved as separate capital. No unavailable previous response is used as a premise. Capital integration is left separate from this research record.

1. The question and the supplied resources

The scientific question is whether acquired knowledge can change the computation needed to learn a later capability. A useful answer must specify the earlier observations, the executable object retained, the later observation interface, and the cost of finding and checking the connection. Retaining more outputs or reusing a convenient latent structure without acquiring it does not answer that question.

What a source row contains

$$X \sim \mathrm{Uniform}(\mathbb{F}_2^n), \quad Z_i = \chi_{c_i}(X)\,\xi_i, \quad \chi_c(x) = (-1)^{c\cdot x}.$$

There are M fixed unknown vectors c_i. Every row contains the actual n input bits and all M channel labels at that same input. The noise signs are independent across channels and rows, independent of X, and have fixed unknown means β_i in (0,1]. Their law is stable across discovery, calibration, and decoding. Positive orientation, the parity model, uniform inputs, and noise independence are supplied assumptions. Finite calibration does not establish them.

The learner receives n, the number of observed channels, a sample callable, finite batch sizes, and rational confidence budgets. It cannot reset a chosen input, request a direct repeated target label, inspect hidden coefficients, or query a latent graph. If a source returns all channels together, all those labels are charged even when only some tasks are eventually admitted.

What old knowledge means

At a later episode the learner retains a finite, frozen dictionary D of q explicit n-bit vectors. Each vector defines an exact parity program. It may have been learned earlier or constructed by a paid XOR of retained programs. D is a finite list, not free access to all 2^n parities. Evaluating an old program is computation; it is not a new observation.

A stored vector need not still carry a correct historical task name. Its execution nevertheless computes exactly the parity described by those bits. The new experiment tests its current relevance. This distinction is the basis of the error-containment result below.

Qualified recovery theorem

For any such history, dictionary, and current task collection, the three-phase procedure in Sections 2–3 acquires a sound graph of task contrasts, calibrates its actual groups, and returns exact programs for the admitted tasks, except with probability at most $\delta_E + \delta_{cal} + \delta_D + \delta_R$. It may return no program. It does not promise that arbitrary tasks supply useful redundancy or that a fixed budget suffices for arbitrarily weak positive biases.

All claims concern exact coefficient recovery on the stated success event. On that event, a returned parity program is correct on every future input, not just on held-out examples. The theorem is ordinary mathematics, independently reconstructed by two helpers; it is not a proof-assistant certificate.

2. Acquiring the operation that aligns observations

Fresh tests discover true contrasts

For every channel pair i < j and candidate d in D, estimate the following moment on A discovery rows:

$$\theta_{ij,d} = E[Z_i Z_j \chi_d(X)]$$
$$= \beta_i\beta_j \text{ if } d = c_i + c_j; \quad 0 \text{ otherwise.}$$

This identity uses uniform-input parity orthogonality and input-independent noise. There are Q = q·choose(M,2) candidates. With $\varepsilon_E = \sqrt{[2\ln(2Q/\delta_E)/A]}$, retain only empirical means larger than ε_E. Hoeffding and a union bound imply that every retained edge is true, except on an event of probability δ_E. Independence between different candidate tests is unnecessary. Weak true edges may be absent.

Each retained edge carries its explicit vector d. Select a root r in every connected component and XOR edge vectors along a spanning tree. On the sound-edge event, the resulting mask is $t_i = c_i + c_r$. Check all remaining edges against these masks. Cycle checks establish internal consistency; their external truth still depends on the statistical event.

Path length does not compound new measurement noise

$$W_i = Z_i \chi_{t_i}(X) = \chi_{c_r}(X)\, \xi_i.$$

The path mask is a computed program, not a product of fresh noisy edge observations. Each aligned channel therefore retains just its own original error ξ_i. All channels in a component now observe one root bit with independent errors. The learner has acquired the correspondence that makes this operation possible; it was not handed a grouping or an anchor assignment.

A target-free numerical calibration

For a discovered component C with $\ell \geq 2$, use R fresh rows to compute:

$$T_C = [(\Sigma_{i\in C} W_i)^2 - \ell] / [\ell(\ell-1)].$$

The unknown root sign disappears on squaring. The statistic has range contained in $[-1/(\ell-1),1]$, width $a_\ell = \ell/(\ell-1)$, and expectation $\rho_C = 2\Sigma_{i<j}\beta_i\beta_j / [\ell(\ell-1)]$. It costs a sum and square per row, not an enumeration of all channel pairs. For h components fixed after discovery, a simultaneous lower bound is:

$$\underline{\rho}_C = \max\{0, T_C - a_\ell\sqrt{[\ln(h/\delta_{cal})/(2R)]}\}.$$

With conditional probability at least $1-\delta_{cal}$, every lower bound is valid. If μ_C is the mean of the channel biases, Cauchy–Schwarz gives $\mu_C^2 \geq \underline{\rho}_C$. The calibration therefore supplies a usable noise bound without learning each channel's rate or observing any true target label.

3. Decoding, finite qualification, and cumulative validity

For an independent decoding row, a wrong majority or tie is contained in $\Sigma\xi_i \le 0$. Markov's inequality, independence, and the arithmetic–geometric mean inequality give, for $\lambda > 0$:

$$P(\Sigma\xi_i \le 0) \le \Pi_i(\cosh\lambda - \beta_i\sinh\lambda)$$
$$\le (\cosh\lambda - \mu_C\sinh\lambda)^{\ell}.$$

Optimize at $\tanh\lambda = \mu_C$, with the noiseless endpoint handled by a limit. On the calibration event the per-row failure probability is bounded by:

$$q_C \le (1-\mu_C^2)^{\ell/2} \le (1-\varrho_C)^{\ell/2}.$$

Choose $N = n + \mathrm{ceil}(\log_2(1/\delta_R))$. Before acquiring the decoding batch, admit any set S satisfying the actually checked risk budget:

$$N\,\Sigma_{C\in S}\,(1-\varrho_C)^{\ell_C/2} \le \delta_D.$$

A union bound then controls every root label in the fresh batch. The common $N\times n$ input matrix fails to have full column rank with probability less than $2^{n-N} \le \delta_R$. One shared GF(2) elimination recovers the selected roots, and $c_i = c_r + t_i$ recovers their tasks. The rank event is shared across all targets; it is not charged separately for each one.

Integer and rational decisions

The implementation replaces logarithms and square roots by conservative integer bounds. For edge discovery take $J = \mathrm{ceil}(\log_2(2Q/\delta_E))$ and $s = \mathrm{ceil}(\sqrt{(2JA)})$; accept only a signed sum strictly exceeding s. For calibration take $K = \mathrm{ceil}(\log_2(h/\delta_{cal}))$ and $u = \mathrm{ceil}(\sqrt{(2RK)})/(2R)$, replacing the real radius by $a_\ell u$. The actual code uses M as a conservative upper bound on h.

$$(1-\varrho_C)^{\ell} \le [\delta_D/(Nh)]^2$$

This squared comparison allocates an equal share to each calibrated component and is exact rational arithmetic. It implies the preceding total-risk gate. Squaring is safe because both sides are nonnegative. Integer bit growth, exponentiation, confidence calculations, and all failed qualification attempts are computation to be paid. The rule is sufficient; it is not an optimal decoder or an impossibility threshold.

No uncharged inheritance of old errors

Condition on any past history, including wrong old task associations. The explicit dictionary is now fixed, so the current moment identity and fresh-data proof still apply. Past errors can remove useful contrasts and reduce coverage, but they do not add an uncharged current error term. A claim that all old and new identifications are correct still uses the sum of their episode budgets. An indefinite sequence needs summable allocations and may require increasing resources.

Freshness matters: the acquired subgraph and its admission decisions are data-dependent. This theorem uses separate discovery, calibration, and decoding batches. Repeatedly inspecting a fixed-sample interval until it passes, or voting on its own selection noise, is not covered.

4. Earning the first executable library

The initial procedure assumes arbitrary parity tasks under the same source model. It has no old dictionary, hidden graph promise, numerical noise rate, supplied rank, or redundancy grouping. It searches for a sound, possibly incomplete collection of short checks.

$$E[Z_e Z_f Z_g] = \beta_e \beta_f \beta_g$$
when $c_e + c_f + c_g = 0$, and 0 otherwise.

Test all choose(M,3) distinct triples using the positive-confidence rule. For a target e, an accepted triple {e,f,g} supplies a possible repair pair {f,g}. Compute an actual matching among these pairs, so the selected pairs use disjoint primitive channels. A deterministic greedy scan suffices; it need not find the maximum matching.

$$V_{e,j} = Z_{f_j} Z_{g_j} = \chi_{c_e}(X)\, \xi_{f_j} \xi_{g_j}.$$

Within this target, the product noises are independent and have positive means $\alpha_j = \beta_{f_j} \beta_{g_j}$. Apply the same aggregate calibration and exact risk gate on fresh data. Use another fresh batch for majority correction and a shared GF(2) solve. Different targets may reuse channels: their errors need not be independent, because simultaneous calibration and decoding use union bounds.

The initial failure bound is $\delta_{checks} + \delta_{cal} + \delta_{decode} + \delta_{rank}$. The learner pays every triple considered, matching construction, calibration, decoding, exact solving, basis selection, and retention. Its retained basis is selected from actually recovered task programs.

A nonvacuous source, without a graph oracle

Let hidden vectors $a_0,\ldots,a_{k-1}$ form an affine independent frame, and let native tasks be the pair parities $c_{uv} = a_u + a_v$. The channels are shuffled. Triangles supply true checks, and each target edge has $k-2$ disjoint two-hop repairs. This proves that useful matchings can exist. The final learner neither receives nor reconstructs the hidden vertex graph; it operates directly on discovered triples.

With homogeneous primitive bias β, a repair vote has bias β^2 and its aggregate correlation is β^4. Two-hop decoding and short parity-check recognition are established methods [1,2]. The new record does not claim their noise dependence as an advance over stronger synchronization algorithms.

Where the bootstrap stops

If the initial task vectors are linearly independent, there are no triple identities. On the sound-check event this procedure acquires none. Small matchings or weakly biased channels can also fail qualification. Declining is a real outcome, not evidence that the tasks are unlearnable. This construction is not a general solver for passive learning parity with noise.

A source with codeword-correlated errors can make every short check pass while preventing error correction. That failure is preserved in Section 8. The existence of many relations does not itself create independent evidence.

5. The complete execution and what it verifies

The final fixture uses n = 48. The source privately constructs 276 pair tasks from 24 hidden vertices. Its independent flip rates vary from 0.01 to 0.02. The learner receives none of those hidden parameters; only n, the observable number of channels, a sample callable, batch sizes, and the confidence budget reach its interface.

Resource	Initial acquisition	Later acquisition
Discovery / calibration / decode rows	512 / 1,024 / 56	512 / 1,024 / 57
Candidate hypotheses	3,466,100 triples	58,170 contrasts
Accepted / rejected hypotheses	2,024 / 3,464,076	210 / 57,960
Native noisy labels	439,392	33,453
Observed input bits	76,416	76,464
Smallest calibrated lower bound	0.7747818588	0.7935151600
Exact output / new rank	276 programs / 23	21 programs / 1
Allocated failure	0.02	0.01

The initial learner found 22 disjoint repair pairs for each target. All 276 qualified. Its decoding bank had 227 raw channel errors and no errors after correction. It selected an acquired rank-23 basis, froze the model, and passed those actual vectors to the next learner.

The next source privately chooses a new vector outside that acquired span, combines it with 21 distinct old anchors, and shuffles their identities. Its flip rates vary from 0.015 to 0.045. The old basis and its finite pair combinations form a 277-vector contrast dictionary. The learner finds one 21-channel component. It corrects all root labels and recovers all 21 task programs, adding one independent direction.

These 21 new true task vectors are linearly independent among themselves. Their output marginal is uniform and has no nonempty internal parity check. The useful operation depends on executing retained functions on the new inputs. This source is a deliberately constructed favorable task sequence, not evidence that natural future tasks will supply the same relations.

Verification and boundaries

Models were written and hashed before hidden coefficients and rates were read by their verifiers. The independent replay reproduced both frozen model JSON files byte for byte and every saved array entry. It confirmed 472,845 native noisy labels and 152,880 observed input bits. Proof reviews checked the mathematical interfaces separately from execution.

The original final run took about 13.17 seconds with Python allocation tracing; the isolated replay took about 13.37 seconds. Those timings are diagnostics, not deployment benchmarks. The implementation uses finite 64-bit input words for this n=48 fixture; the theorem permits ordinary arbitrary-length bit vectors. The stated 0.03 bound is conditional on the ideal source model, not an estimated error frequency from pseudorandom simulations.

6. Full costs and equally informed alternatives

The later theorem pays M(A+R+N) primitive labels and n(A+R+N) input bits whenever decoding is attempted. With explicit dense vectors, direct work includes Aqn bit operations for dictionary evaluation, Aq·choose(M,2) correlation operations, path and consistency calculations, (R+N)nM mask work, and shared elimination. A naive shared solve for s roots costs O(Nn(n+s)) field operations. Packing bits reduces word operations; it does not remove their widths or the observation bill.

Initial discovery costs O(A·choose(M,3)) scalar record-products, or packed equivalents. The final fixture's unpacked triple count is 1,774,643,200. The implementation scans every one of the 3,466,100 candidates. Derived repair votes require additional computation: 6,217,728 calibration XORs and 340,032 decoding XORs were counted in the final bootstrap.

Returning s dense task programs costs ns bits before task identities and bookkeeping. A basis alone does not preserve each dependent task's association: the corresponding coefficients or paths must also be retained. Future execution costs a parity evaluation, O(n) bit work or O(ceil(n/w)) ordinary packed-word work, without old-source calls. Enlarging D increases the discovery search unless a separately earned index or selection method replaces it.

Final-data direct learner

An ordinary comparator retains the exact same final observations. It fits candidates from random full-rank input subsets, shares each GF(2) factorization across all tasks, and selects each task's best candidate using the independent solve bank. Numerical noise rates and hidden graphs are not supplied. The fixed 256-attempt run preserves 64/128/256 states before truth comparison.

All 276 initial programs and all 21 later programs were already exact at the 64-attempt checkpoint. That checkpoint took about 3.50 seconds for the initial tasks and 0.48 seconds for the later tasks. The full 256 attempts were actually run and charged, taking about 14.14 and 1.98 seconds. They scored 70,656 initial and 5,376 later candidates, with 3,956,736 and 306,432 held-out parity evaluations. No new source labels were needed.

An exact null test qualifies every selected function at both checkpoints. For a wrong training-frozen parity, the mismatch count on N fresh uniform inputs is Binomial(N,1/2). Its exact lower-tail probability is compared with δ/(256M); one union bound covers all 256 attempts and both checkpoints. Initial winners had at most five errors against a nine-error admission threshold; later winners had at most five against eleven. Thus the ordinary 64-attempt checkpoint matches exact-program confidence with the same total 0.03 allowance. Certification used no extra labels or search; 33,306 parity rechecks took about 0.016 seconds. This confidence mechanism is also ordinary statistical qualification.

Other controls and the consequence

Earlier retained-data candidate searches learned all tasks in a fixture where the graph rule declined three. An exact Walsh–Hadamard maximum-likelihood learner at n=16 recovered all 120 tasks and was faster than the specialized bootstrap in that fixture. A supplied-graph spectral decoder also succeeded; stronger SDP synchronization is established [2]. Direct repeated-input target access, if available, permits ordinary majority without alignment.

No full-cost advantage follows from these results. The final bootstrap costs about 19,104 labels per acquired independent direction, while its next stage costs 33,453 for one direction. These are different task ensembles, and their ratio is not an intrinsic efficiency measure. The constructive consequence is a way for retained executable functions to enable later acquisition under a specified source, with strong competitors free to use it.

7. What is cumulative, and what extends beyond parity

Define a true graph G_D on the current tasks: i and j are adjacent when their actual contrast c_i+c_j belongs to the finite dictionary. This graph measures a useful interaction between retained knowledge and the new experiment. It is not determined by a representation's dimension or its prediction accuracy alone.

A larger dictionary can join previously disconnected groups, exposing more independent observations of a root. With a common supplied positive bias bound β, merging components decreases the conservative risk expression $N\Sigma_C \exp(-|C|\beta^2/2)$. That is an achievable certificate, not the optimal risk or a lower bound on all learners. The unknown heterogeneous-bias whole-component rule is not necessarily monotone: adding many weak channels can worsen an unweighted vote and its aggregate calibration.

The input coordinates already span every parity. New rank refers only to identified task descriptions relative to the previously identified task library. Mathematical expressivity has not increased, and no input information was manufactured. What has changed is that useful dense programs have been located and can now be executed without repeating the search that identified them.

This gives a complete cumulative loop: discover a program; retain its exact description; test its relevance to a future source; use it to reinterpret current measurements; recover another program. Exact descriptions avoid accumulating approximation error in execution. Statistical claims about task identities still carry their own episode budgets.

An explicit broader reduction

Let $h_1,\ldots,h_k$ be arbitrary exact Boolean programs, and suppose their distinct pair products disagree with probability at least γ under an input distribution. New opaque channels have the form $Y_i = g(X)h_{\pi(i)}(X)\xi_i$, with an unknown injection π and $L \geq 3$. Pair products cancel the arbitrary new target g. A finite correlation search can identify the anchor pairs and their shared endpoint. Correcting by the acquired h programs then yields independent noisy copies of g(X).

With a supplied positive bias bound β, $L \geq 2\beta^{-2}\ln(m/\delta)$ makes an m-example cleaned batch entirely correct except with probability δ. Any passive noiseless learner for g's class can then be used on that batch. This is a reduction to a stated noiseless learner, not a claim that every Boolean target is efficiently learnable. The alignment search, separation qualification, old-program execution, labels, and subsequent training are all paid in the accompanying ordinary proof.

The broader reduction requires exact anchors. Approximate anchors may share the same error c(X): pair products cancel it, while every corrected label becomes g(X)c(X). Majority cannot repair a common functional error. Thus the parity construction's finite exact identification is a substantive reason its cumulative execution is cleaner than repeatedly distilling an approximate predictor.

8. Decisive failures and model boundaries

Changed premise	What the learner can observe	Why the conclusion fails
One shared channel error	Every contrast and pair agreement can look perfect	All aligned votes equal one noisy bit; no independent repetition
Codeword-correlated initial errors	Every triangle check is exact	Every two-hop repair repeats the same corrupted edge
Errors depend on X	High average pair agreement	Hard input regions can retain large majority error
Unpaired input rows	Many related tasks, exact old masks	They remain separate ordinary noisy-parity examples
Approximate common anchor error	Anchor pair matching can be exact	All corrected targets share the same functional corruption
Too few useful checks or channels	A valid but weak certificate	This selection rule declines; another learner can still succeed

Dependence can be invisible to the calibration

If every later channel shares one fresh error ξ, correcting masks gives identical copies of $\chi_{C_r}(X)\xi$. Pair products cancel ξ, so calibration can look ideal. The bundle is equivalent to one ordinary noisy-parity observation per X. In the executed misspecified fixture, qualification passed but the final GF(2) equations were inconsistent and the program returned no task. That one rejection is not a theorem that all such violations will be detected.

An additional counterexample preserves conditional independence given X. Let an input event E have probability τ. Outside E, labels are noiseless; inside E, every channel is an independent fair sign. The average pair correlation is $\rho=1-\tau$. For odd ℓ, the actual majority error is $\tau/2$, while incorrectly applying the constant-bias formula would give $\tau^{\ell/2}$. The gap grows with ℓ. Making inside-E biases small and positive preserves a gap. Input independence or an appropriate uniform conditional guarantee is essential.

Access, certification, and cost are different claims

Exact known masks do not create repeated inputs. With unpaired examples, multiplying by a known mask simply converts each record into an ordinary passive noisy-parity example. This is an exact experiment reduction; any hardness conclusion additionally requires the relevant LPN hardness assumption. It does not establish a lower bound for equally informed learners given the paired data used by our positive construction.

Nor does a failed confidence gate prove an information limit. It is a limitation of a sufficient finite rule. Search failure, extra calibration, aborted solves, and rejected proposals remain in the resource ledger. The evidence preserves these failures rather than rewriting them as successful acquisition.

9. Source-informed selection and preserved capital

Astra retrieval began at **00_START_HERE.txt**, followed by task-first learning, representation-acquisition, and program-synthesis navigation. Governing files were retrieved at commit **72f713bfca2d1de2f869a2ec7c289323e6f0a979**. Seven exact sources, their repository paths, blob identities, and SHA-256 hashes are preserved in the evidence bundle. Navigation cards were used as routes, not substitutes for proofs.

The older ephemeral-procedure construction already learns a shared finite-field map from actual expiring sources and tracks a shrinking affine uncertainty family. Its cumulative acquisition bound charges anchor observations, guards, failed work, algebra, and execution. The guarded-family and cohort constructions already distinguish accepted procedures from guaranteed future coverage and pay fresh qualification or randomized self-correction. The retained-relation construction also distinguishes a discovered invariant from an actually available conditional experiment.

These sources ruled out presenting ordinary representation retention as a new principle. Their failures were equally useful: old raw data can determine a later generic fit; fixed reused guards invalidate confidence; a conditional dimension optimum can lose to a joint query policy; and acquired operator laws can lose to retained Krylov or tomography controls. The present comparison grants conventional learners the same retained data and executable descriptions.

Primary precedents

Noisy short-check recognition is established [1], and disjoint two-hop voting appears explicitly in binary synchronization work that also gives stronger SDP methods [2]. Lifelong metafeature acquisition has formal precedents [3]; OOPS already treats reusable programs and charged incremental search [4]. Those facts guide the comparator and prevent a claim that the ingredients are historically new. The full assembly’s priority remains unsettled.

Separate accessible compiler capital

The earlier signed predictive compiler was independently reconstructed and its four scientific checks replayed. Its source-only acquisition returns an autonomous rational trajectory law with a width-dependent KL guarantee. It uses actual conditional-source access and a retained dictionary that is larger than the proof’s hidden witness. Support-safe escaped forests and normalization corrections remain part of its governing result.

That compiler is valuable separate capital. Joint-law accuracy does not justify arbitrary rare conditional experiments, and recursively applying a fourth-root mismatch bound is not a cumulative improvement theorem. An ordinary retained Bayesian model pool supplies a strong competing reuse rule. Its finite product identities were checked on 768 enumerated datasets. The new investigation keeps these facts and failures without treating them as evidence for the parity construction.

The separate capital archive preserves the accessible p1_opening tree unchanged, including its audits, stronger dyadic acquisition, obsolete predecessor drafts with their original names, and control/memory obstructions. The archive index identifies the governing sources; it does not silently promote every preserved intermediate draft to the final theorem.

10. What remains consequential and unresolved

The proved mechanism values retained knowledge by the new measurements it helps interpret. It supplies one concrete route from an acquired program to another acquired program, even when the later tasks have no output-only parity checks. Its strongest reliability feature is conditional current-episode correctness under arbitrary historical task-label errors, obtained by testing executable behavior on fresh data.

The next substantial advance must improve a load-bearing limitation, rather than merely enlarge a favorable gain ratio. Three openings remain: discover or engineer useful independent task relations at an acceptable native cost; acquire a compact searchable contrast dictionary whose use beats scanning the retained library; or extend the operation to approximate programs while preventing common functional errors from propagating. Each would need a full retained-data comparator and a source model that supplies the promised observations.

Reproduction and evidence map

The companion evidence archive contains the ordinary proofs, exact confidence arithmetic, complete input/output snapshots, frozen models, original failures, source audits, and the independent replay manifest. The principal files are:

sources/UNKNOWN_BIAS_CALIBRATION.md — qualified contrast theorem and initial bootstrap

CUMULATIVE_OPERATION.md — validity conditional on any history and the limits of growth

GENERAL_MASK_ACQUISITION.md — broader exact-Boolean-mask reduction

RESOURCE_BOUNDARIES.md — paired access, correlated noise, and input-dependent noise failures

experiment/fully_qualified_bootstrap.py — complete final executable pipeline

experiment/final_data_comparator.py — equally informed direct-target candidate learner

ROOT_VALIDATION.json — isolated replay, source hashes, exact model/array comparisons

sources/astra_provenance.json — exact older governing source provenance

Run the construction in a fresh copy of the evidence directory; its scripts write results beside themselves. Python 3 and NumPy suffice for the learning experiments. Building this report additionally uses ReportLab and DejaVu fonts. The finite fixture stores X in unsigned 64-bit words; it is not a general large-n software release.

```
python3 experiment/fully_qualified_bootstrap.py
python3 experiment/final_data_comparator.py
python3 experiment/certify_final_candidates.py
python3 validate_complete_pipeline.py
```

The 472,845-label figure belongs to one final pipeline. Earlier development fixtures, strengthened controls, and the independent replay are separate executed research work, preserved in their records. They are not erased or silently included in that operational bill. No practical advantage or universal cumulative-learning claim is inferred from a successful fixture.

References

[1] Karimian, Ziapour & Attari (2012). Parity Check Matrix Recognition from Noisy Codewords. Primary source.

[2] Abbe, Bandeira, Bracher & Singer (2014). Decoding Binary Node Labels from Censored Edge Measurements: Phase Transition and Efficient Recovery. Primary source.

[3] Balcan, Blum & Vempala (2015). Efficient Representations for Lifelong Learning and Autoencoding. Primary source.

[4] Schmidhuber (2004). Optimal Ordered Problem Solver. Primary source.

[5] Astra Research Knowledge: pinned task-first entry; exact governing paths in the evidence manifest. Primary source.